Technical specification template: Uninterruptible Power Supply Systems

OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A5. Generated 2026-11-05 07:50 UTC. Package version 0.1.0.dev0.

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Document control

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1. Scope

This template defines the technical requirements for uninterruptible power supply systems for the OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. It is to be completed and issued by the project. Items in [square brackets] are decisions or data the project must supply. Preliminary values from the Ohmetry reference configuration are shown with their data status and must be replaced or confirmed.

2. Applicable standards

- IEC 62040 series (uninterruptible power systems), including performance classification
- IEC 61000 series (EMC and harmonics)
- IEEE 519 (harmonic limits), if adopted by the project
- Editions and applicability of every standard above are NOT VERIFIED. The project shall confirm the current editions and any national deviations.

3. Service conditions and system data

Item	Requirement or preliminary value	Data status
Nominal LV system	400 V line-to-line, 60 Hz, 3-phase	PRELIMINARY_ASSUMPTION. Utility confirmation is the owner's statement, not independently verified
MV supply	13.8 kV	PRELIMINARY_ASSUMPTION. To be confirmed with the utility
Design ambient temperature	30 degC	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Site altitude	0 m	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Prospective fault level	[To be calculated: short-circuit study pending utility, transformer and generator data]	NOT CALCULATED

4. Equipment ratings

Ratings below are preliminary values from the master configuration unless marked otherwise. They are not manufacturer-confirmed and are not an equipment selection.

Item	Requirement or preliminary value	Data status and note
UPS per power path	2	PRELIMINARY_ASSUMPTION. 2N: each UPS carries the whole critical load after loss of the other path
Rated active power (each)	2,500 kW	UNVERIFIED_PLACEHOLDER
Rated apparent power (each)	2,500 kVA	UNVERIFIED_PLACEHOLDER. Both kW and kVA limits apply
Efficiency (design point)	96 %	UNVERIFIED_PLACEHOLDER. A single point. The manufacturer shall declare the efficiency curve
Input power factor	0.99	UNVERIFIED_PLACEHOLDER
Autonomy	10 min at the design load	PRELIMINARY_ASSUMPTION
Topology, bypass, paralleling	[Project to select]	Project decision

5. Technical requirements

5.1 The UPS shall deliver its rated kW and kVA continuously at the site conditions; overload capability shall be declared but shall not be used to meet continuous loading.

5.2 The manufacturer shall declare efficiency at 25, 50, 75 and 100 per cent load [project to confirm the points].

5.3 The input current harmonic distortion, input power factor, rectifier walk-in time and generator compatibility shall be declared.

5.4 Output voltage and frequency regulation, short-circuit contribution and fault clearing capability shall be declared.

5.5 Battery interface, monitoring, alarms, communication and maintenance bypass: [project to define].

6. Manufacturer data required

The tenderer shall supply the datasheets and values below with its offer, each with the document name and revision. A value without a source document is not accepted. These are also tracked in the manufacturer-data register and are imported into the calculations through the data override mechanism.

Parameter id	Parameter	Datasheet	Current status
equipment.ups_rating	UPS rating	UPS data sheet	UNVERIFIED_PLACEHOLDER
equipment.ups_rating_kva	UPS rating (each), apparent power	UPS data sheet	UNVERIFIED_PLACEHOLDER
equipment.ups_efficiency	UPS efficiency	UPS data sheet	UNVERIFIED_PLACEHOLDER
equipment.ups_input_power_factor	UPS input power factor	UPS data sheet	UNVERIFIED_PLACEHOLDER

derating.ups_ambient_factor	UPS ambient derating factor	Manufacturer derating curve for ambient conditions (ups)	UNVERIFIED_PLACEHOLDER
derating.ups_altitude_factor	UPS altitude derating factor	Manufacturer derating curve for altitude conditions (ups)	UNVERIFIED_PLACEHOLDER
derating.ups_installation_factor	UPS installation derating factor	Manufacturer derating curve for installation conditions (ups)	UNVERIFIED_PLACEHOLDER
generators.ups_rectifier_walk_in_s	UPS rectifier walk-in time	UPS data sheet	MISSING
generators.ups_input_current_thd_pct	UPS input current THD	UPS data sheet	MISSING

7. Inspection and quality requirements

7.1 Review of the datasheet, single-line and control philosophy.

7.2 Inspection of enclosures and power components.

7.3 Routine tests on every unit as required by the applicable standard and this specification.

7.4 Type-test reports or evidence for the design, issued by an accredited or recognised laboratory [project to define acceptable evidence].

7.5 Inspection and test plan (ITP) approved before manufacture, naming witness and hold points.

7.6 Calibration certificates for test instruments.

8. Factory acceptance test (FAT) requirements

8.1 Functional tests: transfer to and from battery and bypass, load steps, overload behaviour within declared limits, efficiency measurement at the declared points.

8.2 Input harmonics and walk-in verification [project to define].

8.3 Witnessing: [owner / independent inspector to attend].

8.4 A FAT procedure shall be submitted for approval before the test; the results shall be issued as a signed report, with deviations and their disposition recorded.

9. Commissioning requirements

9.1 Installation checks, battery connection checks, parameter and alarm checks.

9.2 Load bank and autonomy test together with the battery; generator interaction test.

9.3 Failure tests as part of integrated systems testing.

9.4 Commissioning records shall name the test, the instrument and its calibration, the date and the person, and shall be retained with the verification register.

10. Submittals and documentation

- Datasheets, drawings and type-test reports.
- Inspection and test plan, FAT procedure and report.

- Installation, operation and maintenance manuals.
- Spare parts and special tools lists.
- Warranty terms [project to define].

11. Open items (NOT VERIFIED)

- All ratings in section 4 marked PRELIMINARY or UNVERIFIED are open until confirmed from manufacturer data and the project studies.
- The prospective fault level is NOT CALCULATED until the utility, transformer and generator data are supplied.
- Standards editions and local requirements are NOT VERIFIED.
- This template has not been reviewed by an independent engineer.